

Consider the following polyhedron

$$P = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mid x_1 + x_2 \leq 1, 3x_1 + 10x_2 \leq 15, x_1 \geq 0, x_2 \geq 0 \right\}.$$

Find all possible basic solutions, and identify which ones are feasible.